

We measure it.



Fine particle measuring system

testo 380 – The innovative complete solution

In combination with testo 330-2 LL, the innovative complete solution for solid fuels, oil and gas systems.

Unrestricted TÜV test for the limit value levels 1/2 and according to VDI 4206 Sheet 2

Parallel measurement of fine particles, O₂ und CO

Graphic presentation of all measurement values in real time

Especially economic in operation and maintenance

Effortless handling and easy transport

High-tech in a case: Measurement of all relevant values with only one probe



mg/m³

°C

hPa

O₂

CO/H₂

NO

ΔP

The constantly growing number of solid fuel systems leads to a further increase in the emission of fine particles. The new amendment to the 1. BImSchV prescribes fine particle measurement, presenting chimneysweeps, heating constructors and service technicians with new challenges. The new fine particle measuring instrument testo 380 for the first time allows uncomplicated fine particle measurement directly on site. The completely new measurement method developed by Testo enables easy monitoring and implementation of the fine particle limit values. Combustion systems can be optimized to minimum emissions with the system.

Thanks to the integration of the proven flue gas analyzer testo 330-2 LL as the "command centre" of the fine particle measuring system, not only solid fuel, but also gas and oil systems can be tested and maintained with convenient measurement value presentation in real time.

Overview of the measurement system

The fine particle measuring system testo 380 consists of two system components: the fine particle analyzer testo 380 including fine particle probe, and the testo 330-2 LL as a command centre and a flue gas analyzer. Together, this system offers the highest possible level of compactness, ease of handling and precision in the measurement of solid fuel, oil and gas systems.

Fine particle case

for easy transport of the fine particle measuring system. The entire measurement technology is contained in a convenient case with a weight of only 7.9 kg.

Fine particle probe

transforms a part of the raw gas into measurement gas. The contamination of the system is limited to a minimum, and a highly accurate measurement guaranteed at the same time. The innovative technology makes the fine particle probe compact and easy to use.

Compartment for instruction manual

The instruction manual is stored ready to hand in the lid.

Flue gas analyzer testo 330-2 LL (from version 2006)

The command centre of the system measures not only fine particles, but simultaneously also CO, O₂ and other flue gas parameters. The portable instrument can be removed from the case easily, and used for flue gas analysis on oil and gas systems.

Condensate trap and filters

The condensate trap and several filters prepare the raw gas for flue gas analysis in the testo 330-2 LL.

Storage compartment

for various materials such as the cleaning set.

Fine particle sensor

Thanks to sophisticated technology, Testo has succeeded in making fine particle measurement easy. The fine particle values are displayed in real time, so that the effect of any action taken on the boiler can be traced directly.

Pre-heating stretch

ensures optimum gas temperature, and therefore an extremely accurate fine particle measurement.



Further storage space

e.g. for spare sensor module

Printer

(optionally available) for documentation on site

Mains unit

for testo 330-2 LL



Innovative technology

The fine particle probe

Everything you need for your professional fine particle measurement is contained in Testo's own development, the handy fine particle probe. The probe samples the raw gas directly from the flue gas flow and transports it to the testo 330-2 LL for flue gas analysis.

Simultaneously, the raw gas is mixed with fresh air in the rotation diluter – creating the necessary measurement gas for the fine particle measurement. The fine particle probe is also responsible for the measurement of the flue gas temperature and the flue draught. The probe is equipped with a heating element which ensures a constant temperature of 120 °C, in order that the flue gas does not condense during the measurement. The probe can be quickly and effortlessly stored in the measurement box, and just as easily removed again. Other probes are not necessary for the measurement of fine particles.



The rotation diluter

In order to achieve an especially reliable fine particle measurement, the raw gas is passed through a rotation diluter made of technical ceramics. Thanks to the patented technology, the particle concentration is diluted with the help of a defined quantity of fresh air, so that the contamination of the gas paths and the entire measurement system is reduced to a minimum, and at the same time a precise fine particle measurement takes place. This means the system works without deterioration, cleaning takes place using conventional household cotton buds.



The fine particle sensor

The fine particle sensor measures the mass of the particles contained in the measurement gas. For this purpose, the measurement gas is passed on to the oscillating fine particle sensor through a jet. Depending on the mass of the particles deposited, the oscillation frequency changes, thus allowing the particle mass to be determined. Because this calculation can be carried out at very short intervals thanks to Testo technology*, it is possible to follow the measurement values in the display of the testo 330-2 LL in real time during the entire duration of the measurement. This way, no smoke input is ever missed, any change in the heating boiler and its effects are immediately visible, and the system can be adjusted especially quickly and efficiently.

* several patents pending

Ordering data

testo 380 fine particle analyzer

- Without flue gas analyzer testo 330-2 LL (already owned testo 330-2 LL from version 2006 can be used after a Firmware update)



Part no. 0632 3800

testo 380 fine particle measuring system

- testo 380 fine particle analyzer with fine particle probe and cleaning set
- Flue gas analyzer testo 330-2 LL with mains unit (incl. Bluetooth, H₂-compensated CO cell)
- Modular flue gas probe 300 mm
- Combustion air temperature probe 190 mm



Part no. 0632 3801

Accessories

Flue gas analyzer testo 330-2 LL	Part no.	
Bluetooth testo 330-2 LL flue gas analyzer set with Longlife gas sensors; BLUETOOTH® and H ₂ -compensated CO sensor as well as integrated draught and gas zeroing, incl. rech. battery and calibration protocol; with graphic display	0632 3307 70	

Accessories testo 380	Part no.	
testo 606-2 wood and material humidity meter with integrated humidity measurement and NTC air thermometer incl. protection cap, batteries and calibration protocol	0560 6062	
Combustion air temperature probe, immersion depth 190 mm	0600 9787	
testo 317-3 CO monitor, incl. carrying case with belt clip, headphones, wrist strap, sampler and calibration protocol	0632 3173	
Testo fast printer IRDA with wireless infrared interface; 1 roll thermal paper; 4 AA batteries	0554 0549	
BLUETOOTH printer set with wireless BLUETOOTH interface; incl. 1 roll thermal paper, rechargeable battery and mains unit	0554 0553	
Spare thermal paper for printer, permanent ink	0554 0568	
easyheat PC analysis software, shows measurement in form of diagrams, tables and manages customer data.	0554 3332	
USB connection cable instrument-PC, length 2 m	0449 0047	

Spare parts testo 380	Part no.	
Spare fine particle sensor module	0394 0001	
Spare jet	0394 0002	
Spare particle filter for testo 350 analyzer box (20 pcs.)	0554 3381	
Probe attachment chain	0554 9356	
Probe cleaning brush	0554 0228	

Probes and accessories testo 330-2 LL

Retrofits / spare gas sensors	Part no.
O ₂ sensor for testo 330-1 LL/-2 LL	0393 0002
CO sensor (without H ₂ -compensation) for testo 330-1 LL/-2 LL	0393 0051
CO sensor, H ₂ -compensated, 0 to 8000 ppm for testo 330-1 LL/-2 LL	0393 0101
Spare CO _{low} sensor for testo 330-1 LL/-2 LL	0393 0103
Spare NO sensor, 0 to 3000 ppm for testo 330-1 LL/-2 LL	0393 0151
Upgrade NO-sensor; 0 to 3000 ppm; resolution 1 ppm	0554 2151
NO _{low} spare sensor 0 to 300 ppm, 0.1 ppm, ±2 ppm (0 to 39.9 ppm) ±5% of m.v.	0393 0152
Upgrade COlow-sensor; resolution 0.1 ppm	0554 2103
Retrofit BLUETOOTH® interface	0450 3338



Modular flue gas probes	Part no.
Flue gas probe modular, incl. cone for attachment; thermocouple NiCr-Ni; hose 2.2 m; particle filter; length 180 mm; Ø 8 mm; Tmax. 500 °C; TÜV-tested	0600 9760
Flue gas probe; length 300 mm; Ø 8 mm; Tmax. 500 °C; TÜV approval; probe stop; NiCr-Ni thermocouple; 2.2 m hose and particle filter included	0600 9761
Flue gas probe; length 180 mm; Ø 6 mm; Tmax. 500 °C; probe stop; NiCr-Ni thermocouple; 2.2 m hose and particle filter included	0600 9762
Flue gas probe; length 300 mm; Ø 6 mm; Tmax. 500 °C; probe stop; NiCr-Ni thermocouple; 2.2 m hose and particle filter included	0600 9763
Flue gas probe flexible; length 330 mm; Tmax. 180 °C; short-term 200 °C; bending radius max. 90 °C for measuring at inaccessible points; probe stop; NiCr-Ni-thermocouple; 2.2 m hose and particle filter included	0600 9764

Probe accessories	Part no.
Probe shaft; length 180 mm; 8 mm; Tmax. 500 °C	0554 9760
Probe shaft; length 180 mm; Ø 6 mm; Tmax. 500 °C	0554 9762
Probe shaft; length 300 mm; Ø 8 mm; Tmax. 500 °C	0554 9761
Probe shaft length 335 mm, incl. cone, Ø 8 mm, Tmax 1000 °C	0554 8764
Probe shaft flexible; length 330 mm; Ø 10 mm; Tmax. 180 °C	0554 9764
Probe shaft multi-hole; length 300 mm; Ø 8 mm; for mean CO calculation	0554 5762
Probe shaft multi-hole; length 180 mm; Ø 8 mm; for mean CO calculation	0554 5763
Hose extension; 2.8 m; extension cable for probe and analyzer	0554 1202
Probe stop 8 mm; steel; with spring clamp and handle; Tmax. 500 °C	0554 3330
Probe stop 6 mm; steel; with spring clamp and handle; Tmax. 500 °C	0554 3329

Additional probes	Part no.
Dual wall clearance probe for O ₂ supply air measurement	0632 1260
Gas leak detection probe; 0 to 10000 ppm CH ₄ /C ₃ H ₈	0632 3330
Ambient CO probe, for detecting CO in buildings and rooms; 0 to +500 ppm	0632 3331
Ambient CO ₂ probe (requires connection cable 0430 0143)	0632 1240
Connection cable for ambient CO ₂ probe	0430 0143
Fine pressure probe: highly accurate probe for the measurement of differential pressure and temperature, as well as Pitot tube measurement of flow velocities (see technical data)	0638 0330



Probes and accessories testo 330-2 LL

Combustion air temperature probes		Part no.
Combustion air temperature probe, immersion depth 300 mm	0600 9791	
Combustion air temperature probe, immersion depth 190 mm	0600 9787	
Combustion air temperature probe, immersion depth 60 mm	0600 9797	
Additional temperature probes		Part no.
Mini ambient air probe; for separate ambient air temperature measurement; 0 to +80 °C	0600 3692	
Very fast reaction surface probe	0604 0194	
Connection cable	0430 0143	
Accessories testo 330-2 LL		Part no.
Mains unit international 100-240 V AC / 6.3 V DC for mains operation or battery charging in instrument	0554 1096	
Spare battery 2600 mAh	0515 0107	
Charger for spare battery	0554 1103	
Readout adapter for automatic furnaces testo 330-1/-2 LL	0554 1206	
Adhesive pockets testo 330-1/-2 LL for printout, paper barcode labels; 50 off	0554 0116	
Instrument cleaner, 100 ml	0554 1207	
Smoke tester with oil; soot sheet; for measuring soot in flue gas	0554 0307	
Hose connection set with adapter for separate gas pressure measurement	0554 1203	
Pressure set for testing gas line testo 330-1/-2 LL version 2010	0554 1213	
Differential temperature set; consisting of 2 Velcro probes and temperature adapter	0554 1208	
Spare dirt filter, modular probe; 10 off	0554 3385	
easyheat PC analysis software, shows measurement in form of diagrams, tables and manages customer data.	0554 3332	
Full version easyheat and easyheat.mobile testo 330-1/-2 LL, for the presentation of measurement procedures as diagrams, tables and for managing customer data. Please order USB cable 0449 0047 separately.	0554 1210	
USB connection cable instrument to PC testo 330-1/-2 LL / testo 335	0449 0047	
ISO calibration certificate/flue gas	0520 0003	
Basic system case testo 330-1/-2 LL for analyzer, probes and accessories	0516 3330	
System case with double base	0516 3331	
Tools system case with tools section without content, can be interlocked to basic system case	0516 0329	

Technical data

Measuring range, accuracy, resolution

Meas. range	0 to 300 mg/m ³
Accuracy	acc. VDI 4206-2
Resolution	0.1 mg/m ³ (>5mg/m ³)
Memory	500.000 readings

Other instrument information

Storage and transport temperature	-20 to +50 °C
Operating temperature	+5 to +40 °C
Protection class	IP40
Weight	testo 380: 7.9 kg, testo 330-2 LL: 0.65 kg
Dimensions	475 x 360 x 190 mm
Housing material	ABS
Power supply	via internal mains unit: 100 V AC/0.45 A to 240 V AC/0.2 A (50 to 60 Hz)
Power consumption	max. 100 W

Information fine particle probe

Probe length	270 mm
Probe shaft diameter	12 mm
Probe shaft material	Stainless steel 1.4301
Probe cable length	2.2 m
Integrated elements	Draught measurement, sampling, temperature measurement, probe heating, rotation diluter
Flue gas temperature	max. +500 °C
Probe shaft heating	to +120 °C
Rotation diluter	heated up to +80 °C
Status display	LED, shows warm-up phase and operational readiness